METRIC

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SUPERSEDING
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25 July 2014

# PERFORMANCE SPECIFICATION
# ANTI-SEIZE THREAD COMPOUND, MOLYBDENUM DISULFIDE-PETROLATUM

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification presents requirements for one type of anti-seize thread compound composed of molybdenum disulfide and petrolatum.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8030

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

| | |
|---|---|
| VV-P-236* | Petrolatum, Technical |

* Indicates document is inactive for new design.

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

| | |
|---|---|
| ASTM D217 | Standard Test Methods for Cone Penetration of Lubricating Grease |
| ASTM D2709 | Standard Test Method for Water and Sediment in Middle Distillate Fuels by Centrifuge |
| ASTM D4057 | Standard Practice for Manual Sampling of Petroleum and Petroleum Products |

(Copies of these documents are available from https://www.astm.org.)

SAE INTERNATIONAL

| | |
|---|---|
| AMS-M-7866 | Molybdenum Disulfide, Technical, Lubrication Grade |

(Copies of this document are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Toxicity. The materials shall have no adverse effect on the health of personnel when used for their intended purpose.

3.2 Materials. The physical composition of the thread compound shall be such that the material passes requirements in section 3. The mixture of materials specified in Table I has been known to pass the requirements of this specification but is not mandatory.



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TABLE I. Physical composition.

| Ingredient | Percent by weight | |
|---|---|---|
| | Minimum | Maximum |
| Molybdenum disulfide, SAE-AMS-M-7866 | 48 | 52 |
| Petrolatum, VV-P-236 | Balance | Balance |

3.3 Properties of thread compound. The properties of the thread compound shall be as specified in 3.3.1, 3.3.2, and 3.4 when tested as specified in Table II.

3.3.1 Worked Penetration. The worked penetration range (as measured in units of 0.1 mm) of the thread compound shall be a minimum of 170 and a maximum of 260 when tested as specified in Table II.

3.3.2 Stability. The thread compound shall show no separation when tested as specified in Table II.

3.4 Workmanship. The thread compound shall be free from cakes or lumps and hard, gritty particles. There shall be no separation of the mixture when tested as specified in Table II.

3.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

TABLE II. Chemical and physical requirements and test methods.

| Property | Minimum | Maximum | Test Method or Test Paragraph |
|---|---|---|---|
| Workmanship | Pass | | 4.4.3 |
| Cone penetration | | | ASTM D217 |
| Worked penetration (60 double strokes), 0.1 mm | 170 | 260 | |
| Stability, 1,500 rpm for 30 minutes | Pass | | 4.4.4 |

## 4. VERIFICATION

4.1 Classification of inspection. The examination and testing of the thread compound shall be classified as quality conformance inspection (see 4.1.1).



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4.1.1 Quality conformance inspection. Quality conformance inspection shall consist of all tests from Table II. Samples shall be labeled completely with information that identifies the purposes of the sample, name of product, specification number, lot/batch number (see 6.2), date of manufacture, and date of sampling.

4.2 Sampling. Sampling shall be in accordance with ASTM D4057.

4.3 Method of inspection. Tests shall be performed in accordance with the methods specified in Table II and the contractual requirements (see 6.2). Physical and chemical values apply to the average of the determinations made on the samples.

4.3.1 Inspection. Unless otherwise specified, inspection testing shall be performed in accordance with 4.3.2 and the contractual requirements (see 6.2).

4.3.2 Thread compound. The thread compound shall conform to the requirements for the finished thread compound that shall be determined by appropriate examination and testing in accordance with 4.3.3.

4.3.3 Physical and chemical values. Tests shall be performed in accordance with the applicable methods specified in Table II. Physical and chemical values specified in section 3 apply to the arithmetic average of the determinations made on the samples for those values that fall within any stated repeatability or reproducibility limits of the applicable test method.

4.4 Test procedures.

4.4.1 List of test methods. The tests of this specification shall be conducted in accordance with the applicable test method as listed in Table II and other methods as described in 4.4.2 and 4.4.4.

4.4.2 Inspection requirements. Refer to the contract or purchase order for specific inspection requirements.

4.4.3 Workmanship. The sample thread compound shall be inspected visually and a suitable portion shall be worked with a spatula on a glass surface. After working, the thread compound shall be spread with a straight edge and observed to be free from cakes or lumps and hard, gritty particles. There shall be no separation of the mixture.

4.4.4 Stability. The stability of the thread compound shall be determined by placing 50 grams of the compound in each of the two cone-shaped centrifuge tubes, and centrifuging at 1,500 rpm for 30 minutes. An apparatus that can be used for this test is described in ASTM D2709. Separation shall be defined as droplets or layer of oil appearing on the surface of the compound after centrifuging.



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This compound is particularly suitable for use on military aircraft engine spark plugs and threaded fasteners and fittings at temperatures below 427 °C (800 °F). This compound contains molybdenum disulfide which at higher temperature conditions (538 °C (1000 °F) and above) may induce hot corrosion of fastener or contiguous materials. Accordingly, its use at higher temperatures will be avoided. SAE-AMS2518 thread compound should be used in lieu of products directed by this specification in these high temperature applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Lot/Batch number.
c. Quantity of oil required.
d. Packaging requirements (see 5.1).

6.3 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Safety Data Sheets (SDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, *Shelf-Life Management Program: Program Administration*; and Volume 2, *Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.



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6.6 Subject term (key word) listing.

Petroleum Jelly

Shelf Life

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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## CONCLUDING MATERIAL

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Preparing activity:
Air Force – 68
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